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CONSENT TO PARTICIPATE IN RESEARCH

Piloting Online Learning Tasks to Test for Evidence of Learning and Sleep Benefits

Cohort A

You have been asked to participate in a research study conducted by Cristina Velasquez, PhD from the Department of Media Arts and Sciences at the Massachusetts Institute of Technology (M.I.T.). This study is funded by the MIT-Novo Nordisk AI Postdoctoral Fellows Program and the MIT media lab's women's health initiative.

You were selected as a possible participant in this study because you are in the age range of 60-80 years. We expect 100 people will be in this research study.

*The information below provides a summary of the research. Your participation in this research is voluntary and you can withdraw at any time.

- **Purpose**

The purpose of this study is to determine whether our online learning tasks are appropriately challenging for people in the age range of 60-80 years. This will help us to design future studies better.

- **Study Procedures**

The study will involve **TWO** online learning sessions, one in the **MORNING** and one in the **EVENING** of the **SAME DAY**. We therefore expect that you will be in this research study for about 12 hours in total, although the tasks themselves should take no longer than a combined 40 mins to complete (approximately 30 mins for session 1, and 10 mins for session 2). The online tasks will test your memory, reaction time, and executive function.

- **Risks & Potential Discomfort**

You may find these tasks difficult, repetitive, and/or boring – if this reaches unmanageable levels you may quit the study.

- **Alternatives to Participation**

You can choose not to participate.

Your participation in this research is completely **VOLUNTARY**. You should read the information below, and ask questions about anything you do not understand before deciding whether or not to participate.

- **PURPOSE OF THE STUDY**



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The purpose of this research study is to determine whether our online learning tasks are appropriately challenging for people in the age range of 60-80 years. This will help us to design future studies better.

• PROCEDURES

If you volunteer to participate in this study, we would ask you to do the following experimental procedures:



MORNING: within the first 2 hours after waking up

SESSION 1: LEARNING (Approx. 30 mins)

- **Reaction time task:** you will be asked to press different buttons on your keyboard as quickly and accurately as possible in response to different visual stimuli
- **Executive functioning task:** you will be asked to click targets on your computer screen in a specific order
- **Face-name task:** You will be asked to learn and recall the names of different individuals.

12 HOUR INTERVAL



EVENING: About 2 hours before bed

SESSION 2: TEST (Approx. 10 mins)

- **Reaction time task:** you will be asked to press different buttons on your keyboard as quickly and accurately as possible in response to different visual stimuli
- **Executive functioning task:** you will be asked to click targets on your computer screen in a specific order
- **Face-name task:** You will be asked to recall the names of different individuals.

• POTENTIAL RISKS AND DISCOMFORTS

The tasks may be difficult/frustrating and/or repetitive and boring – if this reaches unmanageable levels you may withdraw from the study. This research may involve risks to you that are currently unforeseeable.



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- **ANTICIPATED BENEFITS**

There are no direct benefits to you from taking part in this research study. The findings from this research may help us to design better learning tasks for people in your age range in future studies.

- **PARTICIPATION AND WITHDRAWAL**

Your participation in this research is completely VOLUNTARY. If you choose to participate you may subsequently withdraw from the study at any time without penalty or consequences of any kind. If you choose not to participate, that will not affect your relationship with M.I.T. or your right to other services to which you are otherwise entitled.

Consequences of Withdrawal

You can stop taking part in this research study at any time; it will not be held against you. If you decide to withdraw, contact the study investigator.

You may also withdraw your consent for the use of data already collected about you, but you must do this in writing to Cristina Velasquez, velas972@mit.edu.

Withdrawal of participation by the investigator

The investigator may withdraw you from participating in this research if circumstances arise which warrant doing so. The decision may be made either to protect your health and safety, or because it is part of the research plan that people who develop certain conditions may not continue to participate.

If you are unable to follow procedures required by the research, you may have to drop out, even if you would like to continue. The investigator, Cristina Velasquez, will make the decision and let you know if it is not possible for you to continue.

- **ALTERNATIVES TO PARTICIPATION**

You may choose not to participate.

- **PAYMENT FOR PARTICIPATION**

If you agree to take part in this research study, we will pay you \$8.00 (\$6.00 for session 1 and \$2.00 for session 2) for your time and effort. Under U.S. federal tax law, non-U.S. citizens are subject to a 30% tax withholding.

If you must drop out because the investigator asks you to or because you have decided on your own to withdraw, you will be compensated in accordance with the payment for participation described above.



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- **FINANCIAL OBLIGATION**

Taking part in this study will not lead to any added costs for you.

- **PRIVACY AND CONFIDENTIALITY**

The following will know that you are a research participant and may inspect your study records:

- Members of the research team which might include outside collaborators not affiliated with MIT.
- Authorized MIT representatives to ensure compliance with MIT policies and procedures.

No information about you, or provided by you during the research will be disclosed to others not listed above without your written permission unless otherwise specified in this consent form, except: if necessary to protect your rights or welfare, or if required by law.

The data will only be linked to your Prolific ID. When the study is completed and the required data retention has passed, all identifiable data will be destroyed.

When results of the research are published or discussed in conferences, data may include aggregate, de-identified, individual-level data, or summary information.

You have the right to review your data, including photographs or recordings, at any time before your completion of this study. When your participation is complete and data analysis has concluded, the research team's ability to honor your request to review your data is limited. All such requests will be reviewed on a case-by-case basis.

- **FUTURE DATA USE**

Your data (such as your information, biospecimens, cell lines, photographs, videos, audios) collected as part of the research, even if identifiers are removed, will not be used or distributed for future research studies.

Traditionally used identifying information about you such as your name, address, phone number, medical record, social security number, your identifiable features or your voices, etc. will be removed before using or distributing for future research. Your samples and information will be available for any research question, such as research aimed at understanding the development and causes of many diseases and conditions or the development of new scientific methods. Your data may be stored for at least 5 years.

- **(*)NEW FINDINGS**



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During the course of the study, you will be informed of any significant new findings (either good or bad), such as changes in the risks or benefits resulting from participation in the research or new alternatives to participation, that might cause you to change your mind about continuing in the study. If new information is provided to you, your consent to continue participating in this study may be re-obtained.

- **(*)EMERGENCY CARE AND COMPENSATION FOR INJURY**

If you feel you have suffered an injury, which may include emotional trauma, as a result of participating in this study, please contact the person in charge of the study as soon as possible.

In the event you suffer such an injury from the direct result of your participation in this study, M.I.T. will provide itself, or arrange for the provision of, emergency transport or medical treatment, including emergency treatment and follow-up care, as needed, and reimbursement for such medical services not covered by your insurance. M.I.T. does not provide any other form of compensation for injury, however, you are not precluded from seeking to collect compensation for injury related to malpractice, fault, or blame on the part of those involved in the research.

You will be financially responsible for any services received for any injuries determined not to be directly related to your participation in the study.

- **(*)IDENTIFICATION OF INVESTIGATORS**

In the event of a research related injury or if you experience an adverse reaction, please immediately contact one of the investigators listed below. If you have any questions about the research, please feel free to contact the following

Principal Investigator: Cristina Velasquez, velas972@mit.edu

Faculty Advisor: Patricia Maes, pattie@media.mit.edu

- **(*)RIGHTS OF STUDY PARTICIPANTS**

You are not waiving any legal claims, rights or remedies because of your participation in this research study. If you feel you have been treated unfairly, or you have questions regarding your rights as a study participant you may contact the Chairman of the Committee on the Use of Humans as Experimental Subjects, M.I.T., Room E25-143B, 77 Massachusetts Ave, Cambridge, MA 02139, phone 1-617-253 6787.



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SIGNATURE OF RESEARCH PARTICIPANT OR LEGAL REPRESENTATIVE
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I have read (or someone has read to me) the information provided above. I have been given an opportunity to ask questions and all of my questions have been answered to my satisfaction. I have been given a copy of this form.

BY CLICKING “I CONSENT”, I WILLINGLY AGREE TO PARTICIPATE IN THE RESEARCH IT DESCRIBES.

IF YOU CLICK “I DO NOT CONSENT” YOU WILL BE SENT BACK TO THE PROLIFIC PAGE AND WILL NOT PARTICIPATE IN THE STUDY DESCRIBED.